LabOS System Architecture

One central server • many PCs • shared knowledge and experimental evidence

Cloud LLM API

Remote model service

Human / operator computer

Browser only • question, review and authorize

CENTRAL LABOS SERVER

ONE CENTRAL ENDPOINT

Scientific reasoning + deterministic control + durable evidence on one server

Knowledge / memory

SEMANTIC KNOWLEDGE

Scientific literature + protocols
Instrument manuals + local code
Calibrations + setup definitions

EXPERIMENT / EPISODIC MEMORY

Prior runs + raw artifacts
Commands, faults + recoveries
Analyses + provenance
Run / setup / capability registry

Retrieve → cite → reuse
Versioned records + configuration hashes

RAG

Save

Main AI agent / control system

Scientific objective → measurement strategy → evidence-based next step

Question + context → Review evidence → Build stage plan → Interpret + refine

Delegate specialist work • compare controls • propose the next scientific stage

Specialist sub-agents

- Literature / protocols
- Planning / sequencing
- Analysis / fitting
- Interpretation / controls
- Troubleshooting
- Code / adapter review

Stage proposal

Qualified evidence

Deterministic execution coordinator

Schema + units | Limits + resource locks | Stage / point barriers | Command / data ledger

Per-PC identity • deadlines • cancellation • qualification • exclusive device ownership

Model router / failover

Preferred: cloud model API
Fallback: local LLM on this server
Triggers: API / internet / quota
Context + permissions preserved

NETWORK • HTTP(S) OVER LAN / VPN

Commands ↓

Evidence ↑

PC-initiated connections • no SSH dependency
Register / heartbeat / poll / results / artifacts

PC A / Optical lab

SuperK + IR spectroscopy

Local PC Agent
Persistent execution worker

Driver adapters + sessions
Journal • watchdog • raw-data outbox

Vendor SDK / USB / LAN

PHYSICAL DEVICES

- SuperK
- IR spectrometer

PC B / Sample lab

Environment + electrical / optical

Local PC Agent
Persistent execution worker

Driver adapters + sessions
Journal • watchdog • raw-data outbox

SDK / VISA / serial

PHYSICAL DEVICES

- attoDRY + transport lock-ins ×2
- BVPE lock-in / chopper / PD
- 730 nm / CLD1000 / 2400

Motorized polarizer / QWP

PC C / Imaging lab

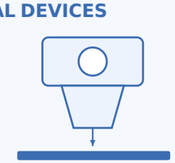
Microscopy + camera acquisition

Local PC Agent
Persistent execution worker

Driver adapters + sessions
Journal • watchdog • raw-data outbox

Camera SDK / USB / LAN

PHYSICAL DEVICES



Microscope / camera

PC D / Motion lab

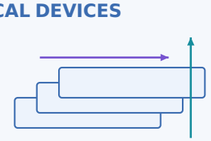
Positioning + sample alignment

Local PC Agent
Persistent execution worker

Driver adapters + sessions
Journal • watchdog • raw-data outbox

USB / serial / Ethernet

PHYSICAL DEVICES



Stages / motors / encoders

PC E / Electrical lab

Source + readout resources

Local PC Agent
Persistent execution worker

Driver adapters + sessions
Journal • watchdog • raw-data outbox

VISA / GPIB / serial

PHYSICAL DEVICES

- Current / voltage source
- Meter / lock-in / digitizer

PC N / Additional lab

Expandable instrument resources

Local PC Agent
Persistent execution worker

Driver adapters + sessions
Journal • watchdog • raw-data outbox

Driver-specific interfaces

PHYSICAL DEVICES

- Instrument / controller
- Detector / sensor

Server: cloud / local LLM routing. Instrument PCs: execution workers; no PC-side LLM required.

A/B retain the supplied setup; C…N illustrate additional hosts. Hardware trigger / reference wiring is separate from HTTP(S).

One experiment → coordinated stages across PCs

Original LabOS system-design figure • scalable deployment view